Topic 4 Computer Software

By the end of this Topic, you should be able to:
1. Define what is meant by computer software;
2. Explain the types and roles of operating systems (Using MS-Windows and its functions);
3. Explain other types of system software;
4. Describe what is meant by application software;
5. Explain major types of application software;
6. Explain the role of middleware and enterprise software; and
7. Describe three methods for acquiring computer software

INTRODUCTION

After learning about hardware let's now discuss about another component called software. The software is an important component of a computer system. Without software, a computer is useless and cannot be used for any meaningful activities.
Computer software can be divided into two classes as shown in Figure 4.1:

1. Application software: A set of computer software developed to perform a specific task or a business process. Examples of application software include word processing, computer aided design, students' information, payroll system, and games.
2. System software: This is the type of software that acts as intermediaries between the hardware and the application software. System software includes operating systems, language translators, utility programs, and performance monitors.

Figure 4.1: Classification of computer software

Computer software is different from computer hardware since it is written instead of manufactured. The process of writing computer software is called computer programming. The person who is responsible for developing computer software is called the software engineer or computer programmer. The process of developing computer software will be discussed in Section 4.7.3.

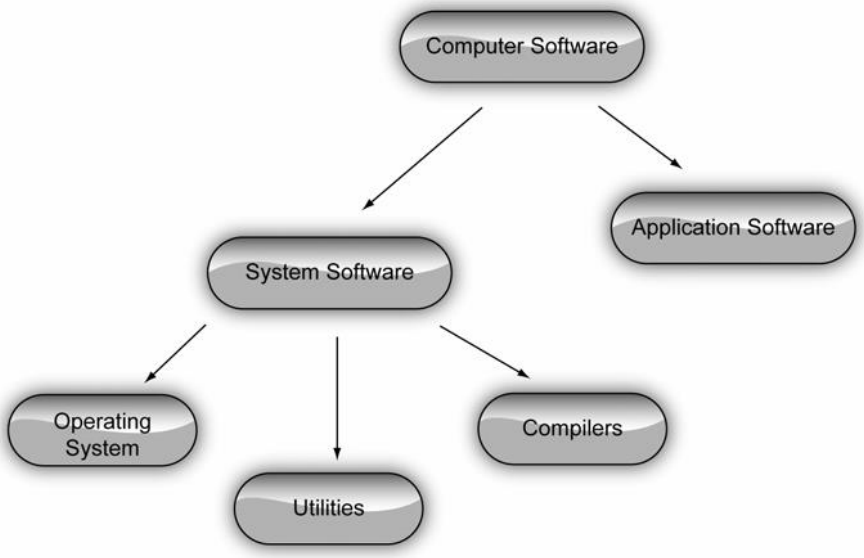


Figure 4.1: Computer software

4.1 OPERATING SYSTEMS

Operating system is the software that supervises the overall operation of a computer system. The main function of an operating system is to provide a virtual machine that is easier to use and hide all the complexity of the underlying computer hardware as shown in Figure 4.2.

This virtual machine environment is:

- Providing an interface between the user and the computer hardware. When a user uses a computer system, he is not really interacting with the computer hardware, but rather he is interacting with the operating system.
- Managing all of the computer resources such as CPU, primary storage, secondary storage, and Input-Output Peripherals.

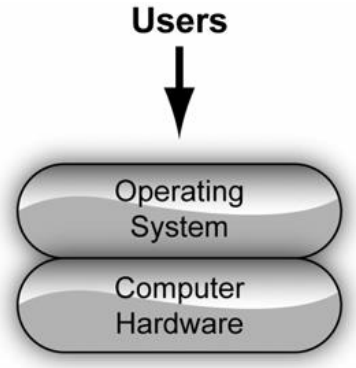


Figure 4.2: Operating System and Computer Hardware

Two most important operating systems are UNIX and Microsoft Windows. UNIX is a dominant operating system on high-end workstations and servers, while Microsoft Windows operating system is widely used on personal computers (PC). The third operating system is LINUX, which is similar to UNIX but is meant for personal computers. Another operating system that is also popular nowadays is the OSX which is the operating system for Macbook computers. OSX is also based on UNIX operating system.

4.1.1 Interacting with Operating System

Users tend to use one of the following interfaces for interacting with computer systems. Each user interface is developed by various professionals to meet different categories of users in computer systems according to their abilities (user-abilities). The available interfaces for users via operating system in this contemporary world include

1) Command Line Interface (CLI),
2) Graphical User Interface (GUI),
3) Touch-based or direct manipulation interface, and
4) Anthropomorphic, voice-based or gesture-based interface.

In CLI user interface, commands are given by typing the text. For example: md simmons

Disk Operating System (DOS)

DOS simple means Disk Operating System. It is software that is loaded when the system is switched on, to provide an environment for other application programs to run. Its function is to organize the running of user programs and transfer of data between the various devices. In addition, it monitors and controls the use of the hardware.

In GUI user interface, commands are given by pointing at icons or menus provided by the system.

4.1.2 Files, Processes, and Virtual Memory

Files and processes are two important resources of a computer system. These two resources, however, are both physical resources but virtual resources created by the operating system.

Files enable us to store information in a computer system in an organized manner. A file encompasses zero (0) or more records treated as unity, or anything on the storage device meaningful to the user. Each file has a file name which is a string of characters that uniquely identify each at a specific storage location or device to avoid duplications. Some of the functions provided by the operating system for file handling are:

- Rename files
- Remove files
- Edit and update files
- Display the content of a file
- Protect files

Since there are a lot of files in a computer system, these files can be organized into directories (in DOS) or folders (in windows). Different operating systems organise directories/folders in different ways. It is possible to have single-level directory, two-level directory or multilevel directory systems as shown in Figure 4.3.

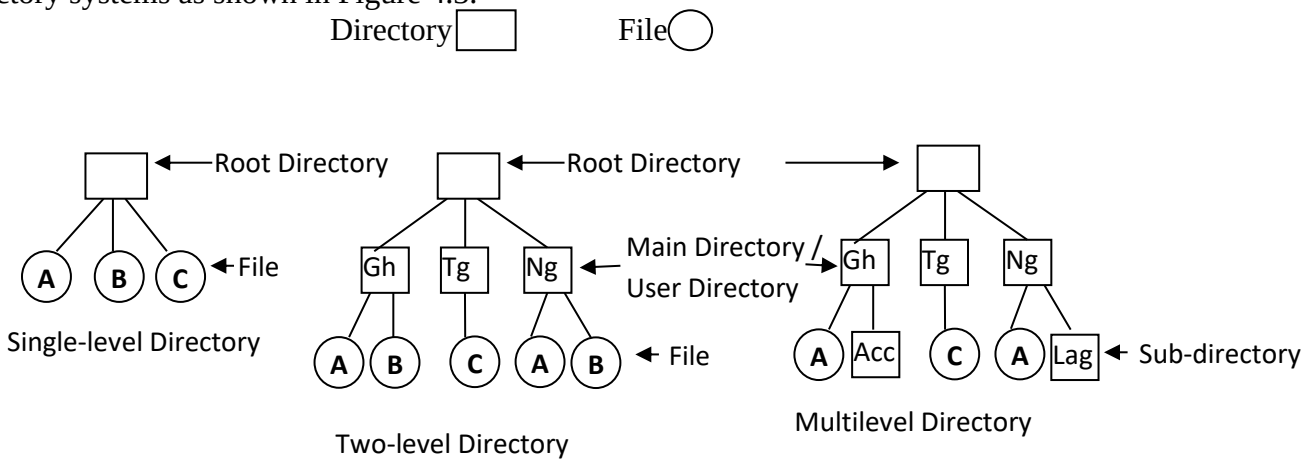


Figure 4.3: Directory levels

FILES

A file encompasses zero (0) or more records treated as a unity or any thing on the disk meaningful to the user. Data is stored in secondary storage on the basis of hierarchies in the context of database. The Figure behind shows file hierarchy chart from top to down. Edit command is used to create file in DOS nowadays.

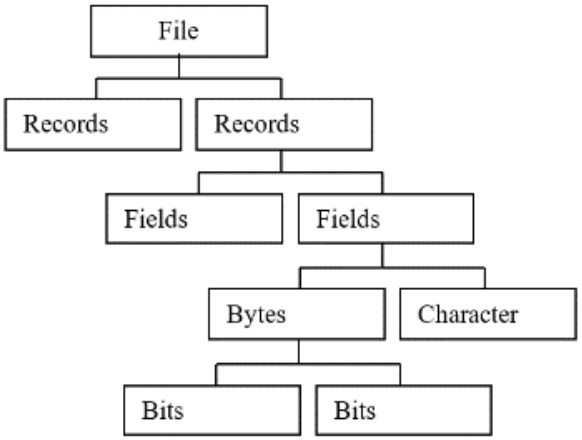


Figure 4.4: File hierarchy

Syntax

C:\>EDIT A [Enter]

1. at the DOS prompt, type edit
2. Leave a space
3. Type the file name (e.g. A)
4. Enter

FOLDERS (DIRECTORIES)

Folder (Directory) is an empty space created purposely to store files as well as sub-folders. The purpose of a directory is to group files and speed searching of files from a storage location. There are two graphical methods for displaying directories, Vertical and Horizontal methods. From the classes of the directories are Root, Main, Sub, Sub-sub, etc. The root folder is any default folder or directory from a storage device, may it be C: for harddisk, E: pendrive, etc. Any directory that comes after the root is called Main Directory. A Sub Directory is also any folder that exists in the main folder, follow by sub-sub directory respectively.

MAKING (FOLDER) DIRECTORY AND SUB-DIRECTORIES

Directory (Folder) is an empty space or cabinet created purposely to store files as well as sub-directories

Keywords: - md
- cd
- cd..
- rd
- cd\

TO MAKE A MAY DIRECTORY

Syntax

C:\>md Gh [Enter]

5. at the DOS prompt, type md
6. Leave a space
7. Type the directory name (e.g. Gh)
8. Enter

CHANGING INTO A DIRECTORY OR WORKING INSIDE A DIRECTORY

Syntax

C:\>cd Gh [Enter]

1. at the DOS prompt, type cd
2. Leave a space
3. Type the directory name (e.g. Gh)
4. Enter

NB: You can follow the steps above to create sub or sub-sub directories

COMING OUT FROM A DIRECTORY

Syntax

C:\GH>cd.. [Enter] C:\GH\ACC>cd\ [Enter]

CASE STUDY_1

Directory graphically shows how your files are stored to speed up searching. You have been contracted by an NGO to store their multinational associated files in your computer. The programs are to store three different political parties from three different countries in Africa.The project must be created on external drive (E:\) with the project tittle as 'africdemocracy' from three countries on your own. Each country must have three political parties & a file of your choice in each. Each political party must also have a file called "A". Illustrate your answer in vertical graphical form of your directory structure. Indicate each level whether Root, Main, etc to suit your folder structure with squares for directories and circles for files.

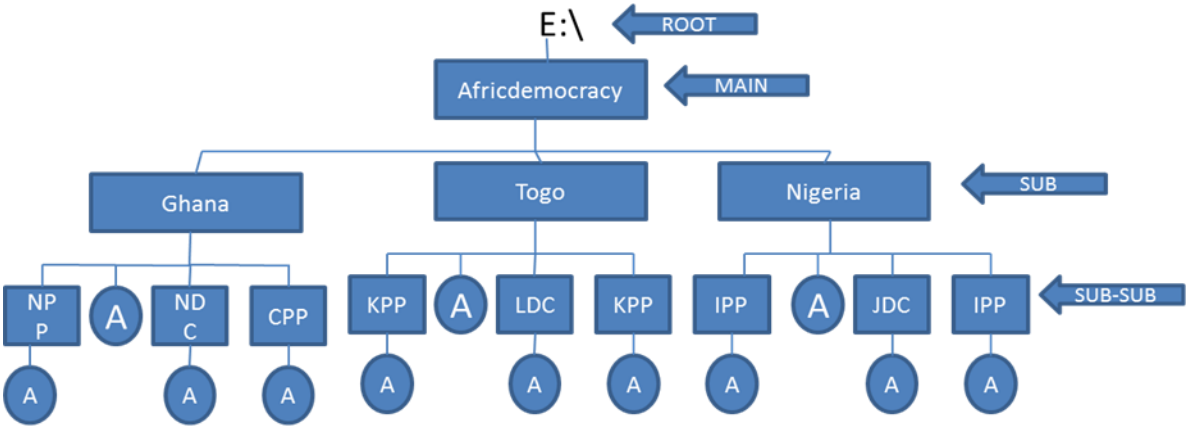


Figure 4.5: Different types of directories structure

PROCESSES

A process is created by executing a program on a CPU. In the earlier version of the operating system, only one process can run a CPU. However, with the introduction of faster hardware, many processes can run on a single CPU at the same time. The management of many processes at one time is called multitasking or multiprogramming. In this environment, a process is executed until an interruption occurs, for example, if the process requests for input. While the input is handled, another process is selected and executed. The task for choosing the next process to run on the CPU is done by a component of the operating system, known as the scheduler. So in this environment, a process can be in various states as shown in Figure 4.5

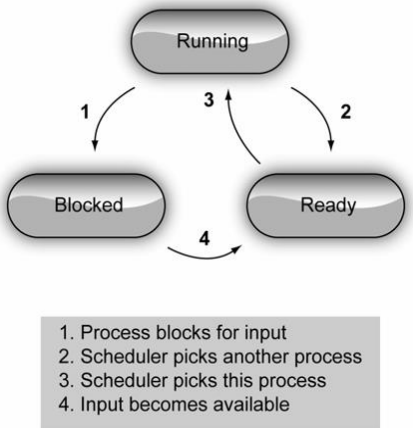


Figure 4.6: Process States

Since the switching among these processes occurs so rapidly, they appear to be executing at the same time. The third type of resources provided by most operating systems is the virtual memory. This virtual memory allows processes to have access to larger memory than is actually provided by the main memory.

Command(s)
An instruction to tell the computer to execute a specific task

CLS
For clearing the screen only at the command prompt or DOS prompt

Syntax
C:\>cls [Enter]

Copy
Use the copy command to copy file(s) from one location to another. This may be from one drive to another, directory to folder or vice versa.

Syntax

C:\>copy simmons E:	[Enter] -	one file copied from c: to e: called simmons
C:\>copy simmons E\GH:	[Enter] -	one file copied from c: to e:\gh called simmons
E:\>copy *.* C:	[Enter] -	all files copied from e: to c:
E:\>copy S*.* C:	[Enter] -	all files begin with S copied from e: to c:

Other commands
Vol
Label
Ver
Date
Time
Tree
Del
Sort

4.1.3 Example 1: UNIX Operating System
In the early days of computers, each computer system has its own operating system. Examples of these operating systems are given in Table 4.1.

Machine	Operating System
IBM 360	System/360
IBM AS/400	OS/400
DEC PDP-11	DEC OS
Honeywell	Honeywell Operating System

Table 4.3: Some old operating systems

However, things have changed after 1970, with the introduction of UNIX and MS-windows 1980 operating system.

4.1.4 Example 2: Microsoft Windows

When IBM introduced IBM PC in 1981, it came together with MS-DOS, an operating system developed by Microsoft Incorporation. MS-DOS was very simple and only supported a single process, which means that only one program can run at one particular time. The user interface to this operating system is based on command line interface.

Microsoft windows is a Graphical User Interface (GUI) program (pictures) which is used in organising the running of user programs and transfer of data between the various devices. It monitors and controls the use of the hardware. In addition, it is a special set of instructions that links hardware and other software programs together. MS-Windows is a multitasking unlike MS-DOS. In the true sense of the word is the ability to allow a single user to perfume more than one task at a time or to run more than one program simultaneously.

Once you learn how to use windows you can use the same basic procedures to control all windows application software. Virtually all application software are designed to take advantage of the windows use the same pull-down menu structures as windows to control processes such as Undo command. In addition windows application software is usually more appealing. Because of the graphical user interface, software written to work with windows also includes many sensible features. For example enables you to use fonts and sizes on the screen as you create a file. Other features like:

- Innovative, easy-to-use features (makes your computer easier to use with new and enhance features)
- Improve reliability: (with new wizards, utilities, and resources that help keep your system running smoothly)
- A faster operating system (a suite of programs designed to optimize your computer's efficiency, especially when used together, in terms of maintenance.
- True web integration
- More entertaining and fun.

To customising your desktop, taskbar, folders, menus, etc.

Icon

It is a diagrammatically representation set of programs in windows. Or can be defined as a smaller picture that represents various types of System (Devices), Application (Programs), Folders (Directories) and Files (Documents). An icon represents a running application or facility whose window has been minimized or closed. That is in order to start an application or facility in windows, the User ought to open into window format

There are three enclose items on the screen (dialog Box, Window, and Message Box)

Dialog box

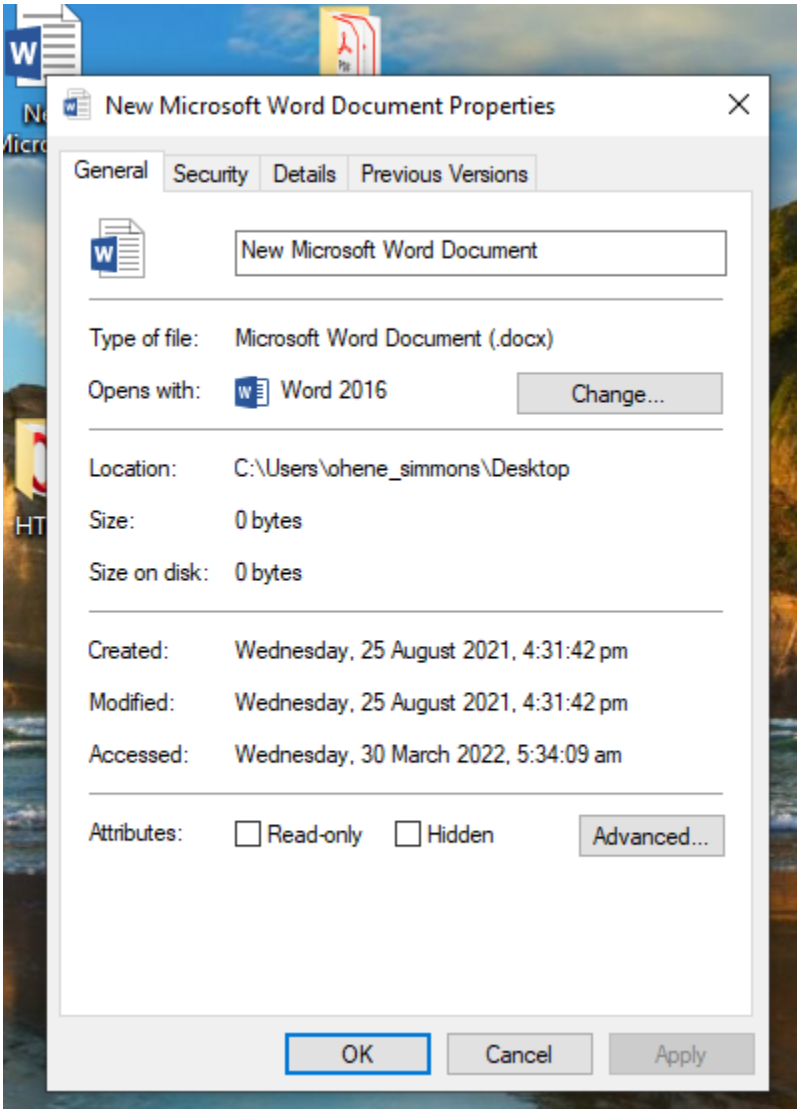


Figure 4.7: Dialog box

Notes

- Windows does not minimize open dialog box
- To open a minimised window, click its button on the taskbar
- To restore your windows to their original state, right-click a blank area on the taskbar, and then click **Undo Minimise All**

Window
A window is and enclosed area on the screen.

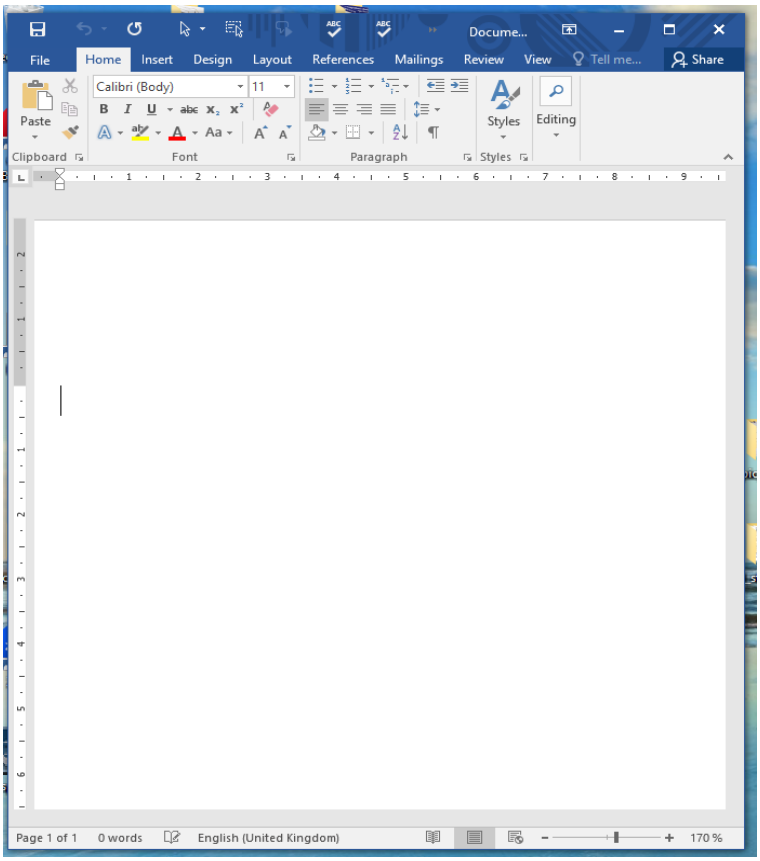


Figure 4.8: Window

Launching a Program

All application programs run in a window. The window is an envelope area on the screen and every window can be moved from one location to another. For example, if you open MS-Excel it will appear in its own window.

Window Types

- Application Window: - contain a running application and a good example is the MS-Excel
- Document Window: - Is said to contain data to be processed in any of the window facilities. An example is a file (book) being processed in MS-Excel
- Folder Window: - It is mainly used for storing programs, files and sub-folders as well.

Other functions

Dialog box pars

How to create and manage icons

Creating window, parts, and controls

Using the windows accessories

- WordPad, Notepad, etc.
- How to manage file(s) such as **S**ave, save **A**s, etc.

Message box

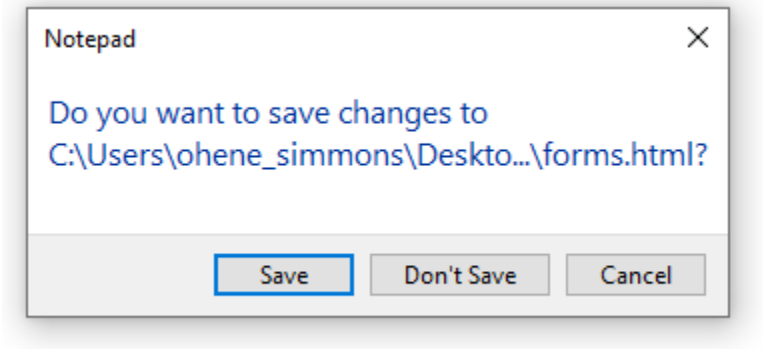


Figure 4.9: Message box

The Recycle Bin

Deleting a file/folder on a disk doesn't mean that it has been deleted or gone permanently. The recycle bin is the temporary location for deleted file(s) and folder(s). Unintentionally deleted files and folders can be restored from the recycle bin. The drawback here is that, deleted files and folders from any movable storage drive do not appear in the recycle bin. Therefore be extra careful when deleting file(s) and folder(s) from any removable storage drive. Deleting anything from the recycle bin means it is permanently gone.

4.1.5 Example 3: Linux Operating System

Linux is another operating system (OS) that can be used for personal computers. It was developed by a Finnish student, Linus Torvalds. The first version of Linux was released in 1991. The major release of Linux was version 1.0, issued in 1994. By this time, a vast amount of UNIX software was ported to Linux, including over 1000 utility programs and X Windows. Many different graphical user interfaces have also been written for Linux. One of them is called K Desktop Environment or KDE (http://www.kde.org).

Linux is very popular due to its business model. It is free software. It can be downloaded from various sites on the Internet. Linux is also considered to be the foundation of Open Source Software. Users may use, copy, modify and redistribute the source and binary codes of Linux freely. The only restriction is that all works derived from Linux may not be sold or redistributed in binary form only.

4.1.6 Operating System for Mobile Devices

A mobile operating system is any an OS designed to suit computing devices such as smartphones, iPads, and smartwatches. Mobile or computing devices normally have a small primary and secondary storage. Thus, they require different types of operating systems. Some of the operating systems for mobile devices are:

- Embedded Linux
- Palm OS
- Windows Mobile
- Symbian OS

Other examples of mobile operating systems are shown in the Figure below.

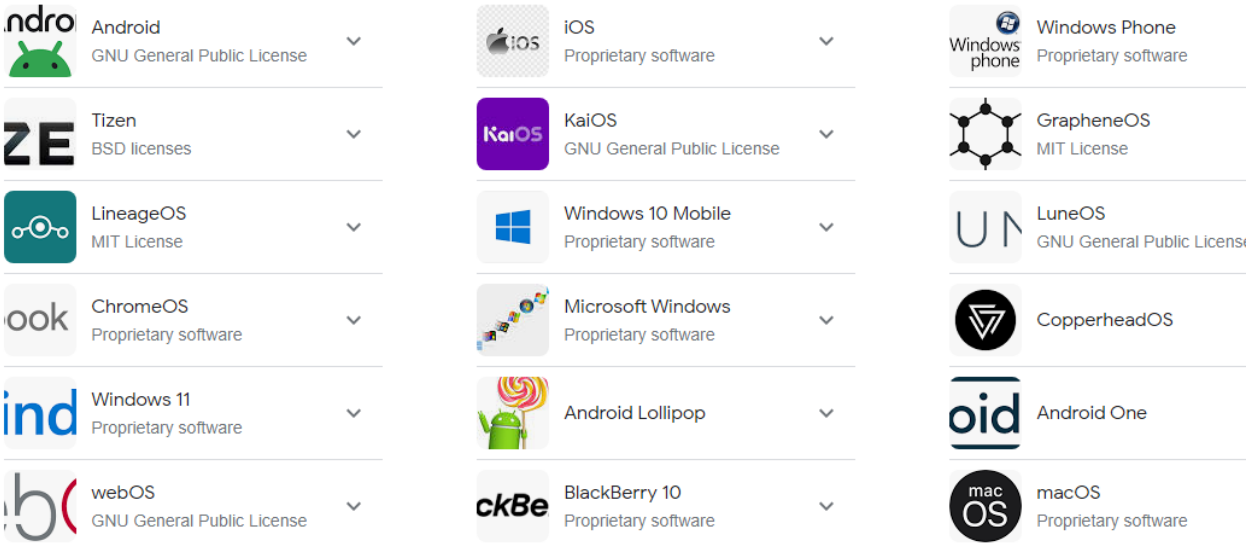


Figure 4.10: Some mobile operating systems

These operating systems are simpler than the normal operating systems and provide much less functionalities. Windows Mobile is a compact operating system developed by Microsoft to be used on devices such as PockectPC, Smartphone, and Portable Media Centers. It is designed to be similar to desktop versions of Microsoft Windows. The first version of Microsoft Mobile appears as Pocket PC 2000 Operating System.

4.2 OTHER TYPES OF SYSTEM SOFTWARE

4.2.1 Language Translator

Apart from operating systems, the most popular type of system software is language translators or commonly known as compilers. The main function of a compiler is to convert programs written in high-level languages into the machine language as shown in Figure 4.11.

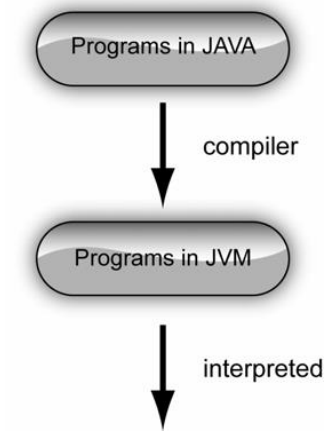


Figure 4.11: Function of a compiler

Machine language is a language that is understood by the CPU. It is based on binary codes. This type of language is also known as the low-level language. Another type of low-level language is called the assembly language. This type of language allows instructions to be represented by using names. High level language is a language that is used by computer programmers to write computer programs.

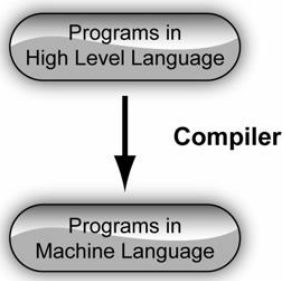
Examples of high level languages are:
C; Pascal; Java; Ada; Visual Basic; Delphi; Haskell; Prolog; LISP

Examples of older high level programming languages include:
Algol; PL/1; POP-11; FORTRAN
High-level programming languages can be divided into various categories as shown below

Categories of high-level languages

Categories	Examples of Languages
Structured Languages	Pascal, C, Ada, Algol, PL-1
Functional Languages	Haskell
Logic Programming Languages	Prolog, LISP
Object-oriented Languages	C++, Java, Python, PHP
Visual Languages	Visual Basic, Bubble, Hypercard, Scratch

Some of the languages, for example Java, are not compiled to the machine language, but to the virtual machine language. The virtual machine for Java is known as Java Virtual Machine (JVM). The advantage of using JVM is that a java program needs to be compiled only once, and it can run on all machines that have the JVM.



4.2.2 Program Editor

Writing a computer program is done by using a software tool called the program editor. This tool enables programmers to create a program file, edit the file and then save the file. Some text editors are small and simple, while others offer a broad and complex range of functionality. For example, two editors provided by UNIX operating systems are vi and Emacs. Microsoft Windows come with the very simple editor known as Notepad.

Some of the common functionalities provided by the editors are:

- ✓ Search and replace;
- ✓ Cut, edit and paste;
- ✓ Undo and redo; and
- ✓ Text formatting.

Many programming languages are now supported by an Integrated Development Environment (IDE). In this development environment, programmers can edit a program file, compile the program and run the program by selecting the right icon or menu provided by the IDE. Some IDE supports more than one language. An example of a multiple-language IDE is Eclipse, that has Java as a base installed language. Eclipse also has plugins for C/C++, Python, Perl, Ruby, Groovy, Fortran, Cobol, and PHP.

4.2.3 Authoring Tools

An Authoring Tool is a program that allows a non-programmer to easily create multimedia or web-based application. The programming features provided by this tool are built in but hidden behind buttons and other tools, so the author does not need to know how to program. Since this tool is mainly designed for developing educational applications, it provides lots of graphics.

Some of the authoring tools for developing multimedia applications are Director, Adobe Director, Authorware Professional, and Multimedia Toolbook.

4.2.4 System Utilities

The main role of system utilities is to support the operation, management, and users of a computer system. Examples of these utilities are:

- ✓ Disk checker: Software that scans for errors in a disk in order to check the integrity of the disk.
- ✓ System performance monitor: Software that monitors the performance of processes in the computer system. Some of the performance measurements are processing time and memory space utilised by the process.
- ✓ Anti-virus software: Software that protects a computer system from virus attack. The main role of anti-virus is t*o identify, neutralise, and eliminate virus.*

4.3 PERSONAL APPLICATION SOFTWARE

There are a large number of application software available covering various types of computer applications. The most popular type of application software is personal application software. The main aim of personal application software is to help users increase their productivity. There are various applications or environments with different naming files. The Table below shows sample software applications and their associated attributes.

Generic Name	Example	File Name	Element (consist of)
Word processing	MS-word	Document	Pages
Desktop publishing	Corel Draw, MS-Publisher	Drawing, Publisher	pages
Spreadsheet	MS-Excel, Lotus	Book	sheets
Presentation	MS-Power Point	Presentation	Slides
Websites	Dreamweaver, HTML	Websites	webpages

4.3.1 Document Preparation Software

Most documents are now prepared by using computers. There are a number of software applications that can be used for this purpose. These applications can be classified into:

- ➢ Word processing software: A type of software that allows user to edit text. One of the earliest word processing software is Wordstar. Most people are now using Microsoft Word and WordPerfect.
- ➢ Text formatter: A type of software that will accept a file prepared by using an editor and then format the file based on the markup provided in the file. Two most popular text formatters are nroff and LaTEX.
- ➢ Desktop publishing: This is a type of software that can produce high quality camera-ready documents. An example of this software is PageMaker and Microsoft Publisher.

4.3.2 Spreadsheet

Spreadsheets software is a type of computer software that allows users to enter numbers into an electronic ledger sheet. This software also provides facilities for users to enter formula to obtain a calculated answer. The first spreadsheet software was VisiCalc which was created by Harvard Business School student by the name of Daniel Bricklin in 1978.

4.3.3 Data Management

Data management software is a type of software that provides facilities for storage, retrieval and manipulation of related data for personal application. An example of a popular data management software is Microsoft Access.

4.3.4 Presentation Software

Presentation software is the application software that is normally set for seminars or lectures. An example of presentation software is Microsoft PowerPoint.

4.3.5 Computer-Aided Software

Computer aided design software is the application used by designers in designing and building production prototypes in software and testing them as computer objects.

4.3.6 Software Suites

Personal application software is normally packaged into software suites. An example of a software suite is Microsoft Office which is a collection of software such as Microsoft Word, Microsoft Excel, and Microsoft PowerPoint. Another popular software suite is OpenOffice, which is a free cross-platform office application suite. It is available for a number of different operating systems and supports over 45 languages. It was originally known as StarOffice, an office suite developed by StarDivision. This company was acquired by Sun Microsystems in August 1999.

4.4 GROUPWARE

Groupware allows people to communicate and share information. This type of software can be divided into two categories: computer supported cooperative work and computer supported social computing.

4.4.1 Communication Tools

Electronic communication tools send messages, files, data, or documents between people and hence facilitate the sharing of information. Examples include:

- Synchronous conferencing;
- e-mail;
- Faxing;
- Voice mail; and
- Wikis.

4.4.2 Conferencing Tools

Electronic conferencing tools facilitate the sharing of information, but in a more interactive way. Examples include:

- Internet forums (also known as message boards or discussion boards) · a virtual discussion platform to facilitate and manage online text messages.
- Online chat- a virtual discussion platform to facilitate and manage real-time text messages.
- Instant Messaging.
- Telephony _ telephones allow users to interact.
- Videoconferencing _ networked PCs share video and audio signals.
- Data conferencing _ networked PCs share a common whiteboard that each user can modify.
- Application sharing _ users can access a shared document or application from their respective computers simultaneously in real time.
- Electronic meeting systems (EMS) _ originally these were described as "electronic meeting systems," and they were built into meeting rooms. These special purpose rooms usually contained video projectors interlinked with numerous PCs; however, electronic meeting systems have evolved into web based, any time, any place systems that will

accommodate "distributed" meeting participants who may be dispersed in several locations.

4.4.3 Collaborative Management Tools

Collaborative management tools facilitate and manage group activities. Examples include:

- Electronic calendars (also called time management software) - schedule events and automatically notify and remind group members
- Project management systems - schedule, track, and chart the steps in a project as it is being completed
- Workflow systems - collaborative management of tasks and documents within a knowledge-based business process
- Knowledge management systems - collect, organise, manage, and share various forms of information
- Prediction markets - let a group of people predict together the outcome of future events
- Extranet systems (sometimes also known as 'project extranets') _ collect, organise, manage and share information associated with the delivery of a project (eg: the construction of a building)
- Social software systems - organise social relations of groups
- Online spreadsheets - collaborate and share structured data and information

4.4.4 Social Computing Tools

The social computing tools are the social media platforms that support trusted and untrusted users across the world to interact. The social media platforms are grouped into two, namely; social media platforms and social networking platforms. The social media platforms are those that are opened to the public to participate without any restriction. Examples of social media platforms or groupware are Facebook, Facebook, and Friendster. Facebook is a free-access website is owned and operated by Facebook, Inc. It was created by Mark Zuckerberg while he was a student at Harvard University. Users can join networks to connect and interact with other people. People can also add friends and send them messages, and update their personal profile to notify friends about themselves. The Friendster site was founded by Jonathan Abrams in March 2002. Friendster was developed for networking individuals in virtual communities.

The social networking platforms are those for only trusted group of users to share information. Examples include WhatsApp and telegram.

4.5 MULTIMEDIA SOFTWARE

Multimedia is a combination of a variety of media (text, audio, video, animation, graphics and images). Multimedia software is a type of software that enables multimedia presentation of information. One example of multimedia software is multimedia-based courseware.
A lot of computer games are now available in the market. Most of these computer games are multimedia-based.

4.6 MIDDLEWARE AND ENTERPRISE SOFTWARE

Enterprise software refer to those applications that are important for the operation of an organisation. The most important application software for an organisation is the computer-based information systems. This system is application software that supports the process of storing, analysing, and dissemination of information for a specific purpose. Most of the enterprise software are complex and

involve a lot of application modules running on heterogeneous platforms. To enable all these application modules to be able to be linked together, there is a need for special software. This type of software is called the middleware.
Computer platform refers to the type of computer hardware and operating system being used.

Heterogeneous platform is a collection of computer systems that involves variety of computer hardware and operating systems. One of the major problems in communication between application modules running on heterogeneous platforms is that each module may use different data format.

4.7 METHODS FOR SOFTWARE ACQUISITION

The process of software acquisition must be started with requirement analysis. During this analysis, the exact requirement of the software will be identified and a software specification document will be produced. We then have a number of options:
(a) Search for publicly available freeware or shareware;
(b) Search for commercial off-the-shelf software (COTS); and
(c) Search for similar software that is currently being used by other organisations.

Once the software is identified, it needs to be evaluated to ensure that it really satisfies the requirement set in the software specification document. If it is not, some customisation process may need to be done. Otherwise, we have to adopt the next possible option that is to develop the software from the beginning.

4.7.1 Freeware and Shareware

Freeware is software that is given for free. This type of software can be divided into two groups: freeware that is available in binary codes and open-source software. Examples of open-source software are:

- GNU suite of software, developed by Free Software Foundation;
- Linux Operating System;
- Apache Web Server;
- Sendmail SMTP;
- Perl programming language;
- Netscape Mozilla web browser; and
- Sun's StarOffice application suites.

Shareware is similar to freeware, except that the developers of the software expect a little financial contribution from the users.

4.7.2 Commercial Off-the-shelf software (COTS)

This is the type of software that is sold by the software vendors. In order to protect their software or products from being copied and distributed illegally, these types are normally copyrighted and thus, they are protected by the copyright laws. The enforcement of the copyright laws is done by Software Publishers Association (SPA). To use copyrighted software legally, a user must purchase a license from the software vendor. Some of these licenses require users to pay for the licenses annually.

Another issue related to COTS is software upgrade. Software vendors always revise their software and sell new versions of the software. In most cases, this implies that users must acquire new license to be able to use the new version of the software.

The relationship between Internet, Extranet, and Intranet

Internet: - it is a network of networks. There has been a dramatic growth in the use of Internet for banks in Ghana communication. The Internet provides an effective, immediate and powerful method for banks to communicate all sorts of issues. This exposes all the banks to the security risks that go with the connection to the Internet. This Internet is really just a backbone connection that enables every computer in the world to connect to every other computer. This gives criminals a direct means of reaching any and every bank information in Ghana that is connected to the Internet. Though there is a firewall that provides some security measures to prevent unauthorised users for accessing other confidential or private information. The relationship between Internet, Firewall, Extranet, and Intranet that allows communication between millions of connected computers worldwide is shown in the Figure ##

Extranet: - It is a private network that uses Internet protocols, network connectivity, and possibly the public telecommunication system to securely share part of an organization's information or operations with suppliers, vendors, customers or other business to do remittances to and from overseas, e.g. Western Union money Transfer. A customer can withdraw or deposit money at any branch of the bank without going to the parent branch.

Intranet: - It is a private network within a single company like Ghana Commercial bank using the Internet technologies or standard to enable employees anywhere in within the firm to share information such as memos. Financial managers can now communicate with their staff effectively to perform routing activities via intranet.

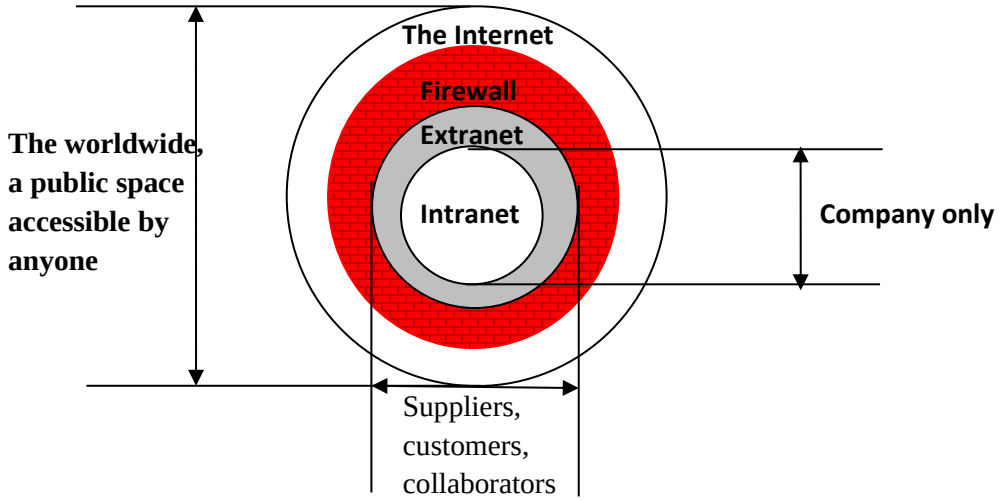


Figure ##: a conceptual diagram among Internet, Firewall, Extranet, and Intranet

Using firewall to prevent criminals

A firewall helps to keep your computer more secure. It restricts information that comes to your computer from other computers, giving you more control over the data on your computer and providing a line of defence against people or programs (including viruses and worms) that try to connect to your computer without invitation.